

CEE 260/MIE 273: Probability and Statistics in Civil Engineering

M1c: Case studies and experiments

Prof. Oke

UMassAmherst

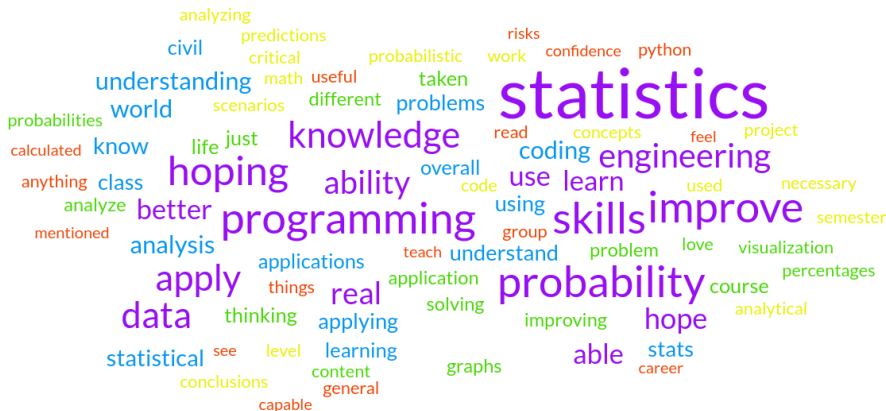
College of Engineering

September 15, 2026

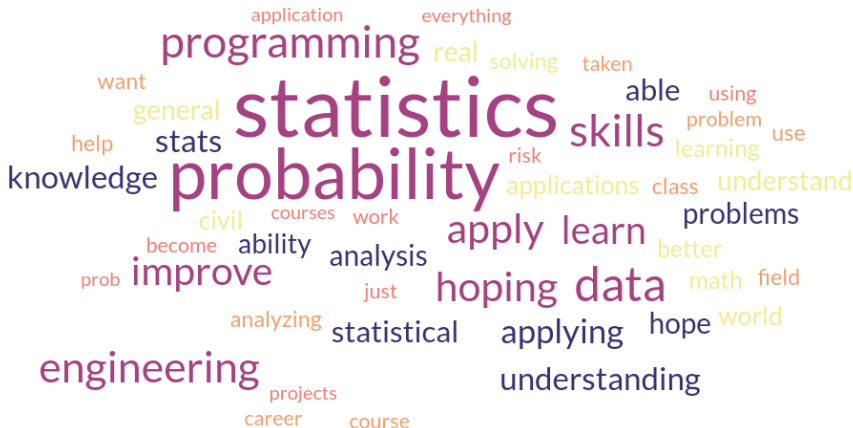
Outline

- ① Recap: Quantiles and boxplots
- ② Colab Examples
- ③ Outlook

Their expectations (Fall 2025)



Your expectations (Fall 2026)



Quantiles

Quantiles

Quantiles are cutoff points that partition an ordered sample or dataset into equal-sized groups.

Quantiles

Quantiles are cutoff points that partition an ordered sample or dataset into equal-sized groups.

- A **median** splits a sample into two: m

Quantiles

Quantiles are cutoff points that partition an ordered sample or dataset into equal-sized groups.

- A **median** splits a sample into two: m
- Two **terciles** split a sample into 3 groups: T_1, T_2

Quantiles

Quantiles are cutoff points that partition an ordered sample or dataset into equal-sized groups.

- A **median** splits a sample into two: m
- Two **terciles** split a sample into 3 groups: T_1, T_2
- Three **quartiles** split a sample into 4 groups: Q_1, Q_2, Q_3

Quantiles

Quantiles are cutoff points that partition an ordered sample or dataset into equal-sized groups.

- A **median** splits a sample into two: m
- Two **terciles** split a sample into 3 groups: T_1, T_2
- Three **quartiles** split a sample into 4 groups: Q_1, Q_2, Q_3
- Four **quintiles** split a sample into 5 groups: QU_1, QU_2, QU_3, QU_4

Quantiles

Quantiles are cutoff points that partition an ordered sample or dataset into equal-sized groups.

- A **median** splits a sample into two: m
- Two **terciles** split a sample into 3 groups: T_1, T_2
- Three **quartiles** split a sample into 4 groups: Q_1, Q_2, Q_3
- Four **quintiles** split a sample into 5 groups: QU_1, QU_2, QU_3, QU_4
- ...

Quantiles

Quantiles are cutoff points that partition an ordered sample or dataset into equal-sized groups.

- A **median** splits a sample into two: m
- Two **terciles** split a sample into 3 groups: T_1, T_2
- Three **quartiles** split a sample into 4 groups: Q_1, Q_2, Q_3
- Four **quintiles** split a sample into 5 groups: QU_1, QU_2, QU_3, QU_4
- ...
- Ninety-nine **[per]centiles** split a sample into 100 groups: $P_1, \dots, P_{99}$

Quantiles (cont.)

Quantiles (cont.)

Certain quantiles are equivalent to others:

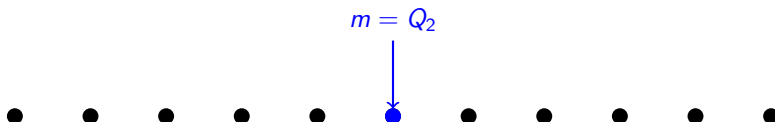
Quantiles (cont.)

Certain quantiles are equivalent to others:



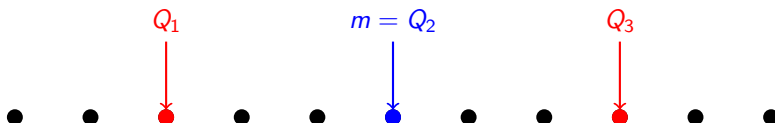
Quantiles (cont.)

Certain quantiles are equivalent to others:



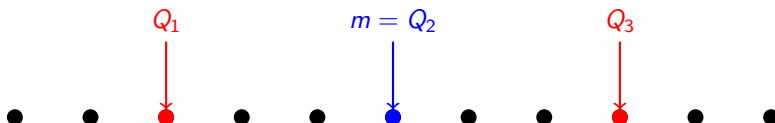
Quantiles (cont.)

Certain quantiles are equivalent to others:



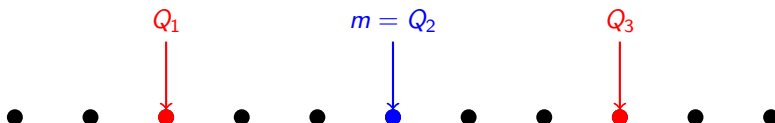
Quantiles (cont.)

Certain quantiles are equivalent to others:



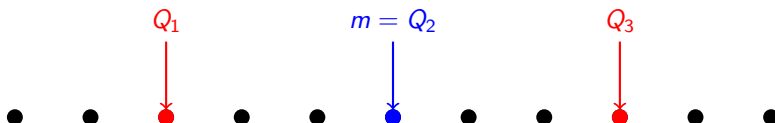
Quantiles (cont.)

Certain quantiles are equivalent to others:



Quantiles (cont.)

Certain quantiles are equivalent to others:

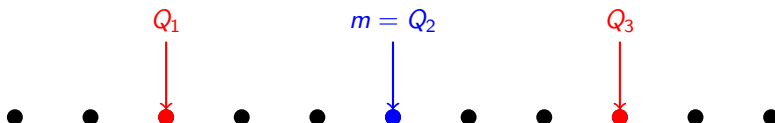


- The median is the second quartile Q_2

Sextiles ($S_1, S_2, \dots$) partition a distribution into 6 equal groups.

Quantiles (cont.)

Certain quantiles are equivalent to others:

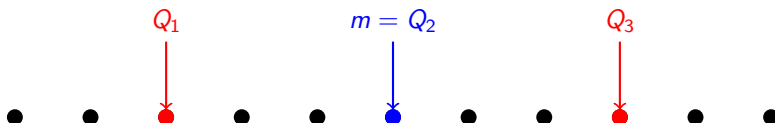


- The median is the second quartile Q_2
- The 25th percentile is equivalent to the first quartile Q_1

Sextiles ($S_1, S_2, \dots$) partition a distribution into 6 equal groups.

Quantiles (cont.)

Certain quantiles are equivalent to others:



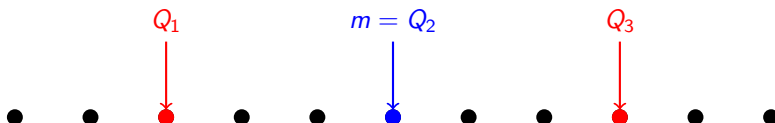
- The median is the second quartile Q_2
- The 25th percentile is equivalent to the first quartile Q_1

Sextiles ($S_1, S_2, \dots$) partition a distribution into 6 equal groups.

- ① How many sextiles are there?

Quantiles (cont.)

Certain quantiles are equivalent to others:



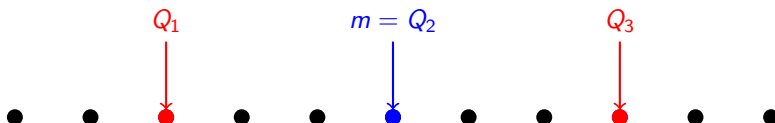
- The median is the second quartile Q_2
- The 25th percentile is equivalent to the first quartile Q_1

Sextiles ($S_1, S_2, \dots$) partition a distribution into 6 equal groups.

- ① How many sextiles are there?
- ② The second sextile S_2 can be expressed as which tercile?^a

Quantiles (cont.)

Certain quantiles are equivalent to others:



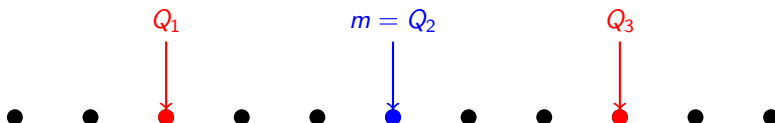
- The median is the second quartile Q_2
- The 25th percentile is equivalent to the first quartile Q_1

Sextiles ($S_1, S_2, \dots$) partition a distribution into 6 equal groups.

- ① How many sextiles are there?
- ② The second sextile S_2 can be expressed as which tercile?^a

Quantiles (cont.)

Certain quantiles are equivalent to others:



- The median is the second quartile Q_2
- The 25th percentile is equivalent to the first quartile Q_1

Sextiles ($S_1, S_2, \dots$) partition a distribution into 6 equal groups.

- ① How many sextiles are there?
- ② The second sextile S_2 can be expressed as which tercile?^a

Answers: Q1: There are 5 sextiles; Q2: $S_2 = T_1$ (first tercile)

^aRecall that a tercile splits a sample into 3 equal groups.

Using boxplots

¹Figure source: <https://towardsdatascience.com/understanding-boxplots-5e2df7bcd51>

Using boxplots

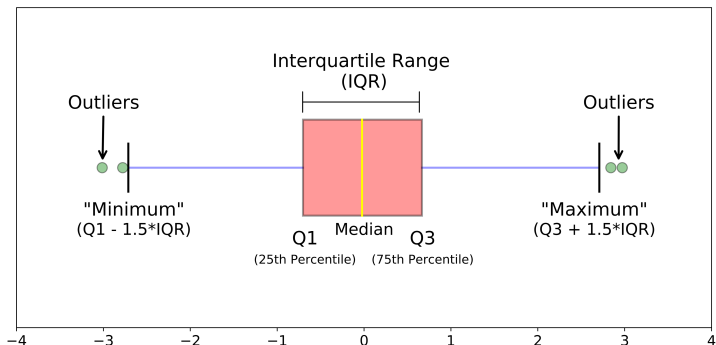
A boxplot¹ displays the distribution of data

¹Figure source: <https://towardsdatascience.com/understanding-boxplots-5e2df7bcd51>

Using boxplots

A boxplot¹ displays the distribution of data

- Useful for identifying outliers

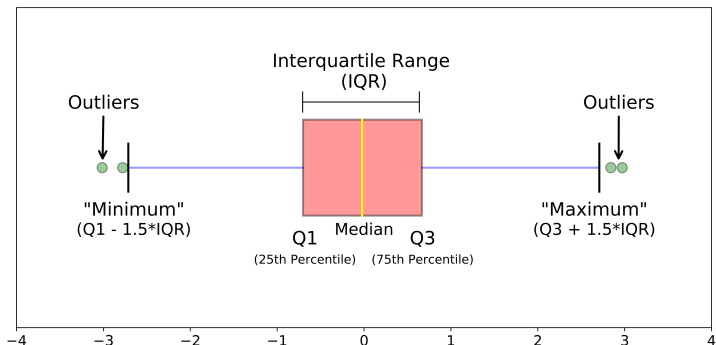


¹Figure source: <https://towardsdatascience.com/understanding-boxplots-5e2df7bcd51>

Using boxplots

A boxplot¹ displays the distribution of data

- Useful for identifying outliers
- Efficient for comparing multiple datasets

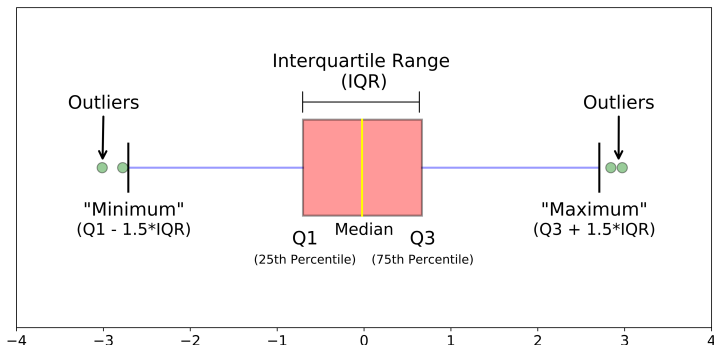


¹Figure source: <https://towardsdatascience.com/understanding-boxplots-5e2df7bcd51>

Using boxplots

A boxplot¹ displays the distribution of data

- Useful for identifying outliers
- Efficient for comparing multiple datasets
- The lines indicating the “maximum/minimum” points (excluding outliers) are called *whiskers*



¹Figure source: <https://towardsdatascience.com/understanding-boxplots-5e2df7bcd51>

Google Colaboratory

Google Colaboratory

We will begin our introduction to Python via basic statistical analyses using the Google Colab platform (<https://colab.google/>).

Google Colaboratory

We will begin our introduction to Python via basic statistical analyses using the Google Colab platform (<https://colab.google/>).

Notebooks

Notebooks

Today, we will use the following Notebooks:

Notebooks

Today, we will use the following Notebooks:

- `m1-blind-stork.ipynb` <https://colab.research.google.com/drive/1dhNi0bMTetlWCozepK24Qk2LElJLtZSt?usp=sharing>
- `m1-usa-housing.ipynb` https://colab.research.google.com/drive/1onpllyTzNuo09op89WDR4ft7o_U-eJVL?usp=sharing

Bonus Example

Bonus Example

Example 1: Walking cadence

Bonus Example

Example 1: Walking cadence

In the article “Can We Really Walk Straight?” (*Amer. J. of Physical Anthropology*, 1992: 19–27) reported on an experiment in which each of 20 healthy men was asked to walk as straight as possible to a target 60m away at normal speed.

Bonus Example

Example 1: Walking cadence

In the article “Can We Really Walk Straight?” (*Amer. J. of Physical Anthropology*, 1992: 19–27) reported on an experiment in which each of 20 healthy men was asked to walk as straight as possible to a target 60m away at normal speed.

Consider the following observations on cadence (number of strides per second):

Bonus Example

Example 1: Walking cadence

In the article “Can We Really Walk Straight?” (*Amer. J. of Physical Anthropology*, 1992: 19–27) reported on an experiment in which each of 20 healthy men was asked to walk as straight as possible to a target 60m away at normal speed.

Consider the following observations on cadence (number of strides per second):

.95 .85 .92 .95 .93 .86 1.00 .92 .85 .81 .78 .93 .93 1.05 .93 1.06 .96 .81 .96

Bonus Example

Example 1: Walking cadence

In the article “Can We Really Walk Straight?” (*Amer. J. of Physical Anthropology*, 1992: 19–27) reported on an experiment in which each of 20 healthy men was asked to walk as straight as possible to a target 60m away at normal speed.

Consider the following observations on cadence (number of strides per second):

.95 .85 .92 .95 .93 .86 1.00 .92 .85 .81 .78 .93 .93 1.05 .93 1.06 .96 .81 .96

Summarize the data; interpret and discuss.

To be assigned: Iris dataset

We will explore the properties and applications of boxplots using the following datasets:

To be assigned: Iris dataset

We will explore the properties and applications of boxplots using the following datasets:

- Popularized in Ronald A. Fisher's classic 1936 paper, "The Use of Multiple Measurements in Taxonomic Problems" <https://onlinelibrary.wiley.com/doi/10.1111/j.1469-1809.1936.tb02137.x>

To be assigned: Iris dataset

We will explore the properties and applications of boxplots using the following datasets:

- Popularized in Ronald A. Fisher's classic 1936 paper, "The Use of Multiple Measurements in Taxonomic Problems" <https://onlinelibrary.wiley.com/doi/10.1111/j.1469-1809.1936.tb02137.x>
- Data collected by Edgar Anderson on various measurements of 3 species of Iris flowers

To be assigned: Iris dataset

We will explore the properties and applications of boxplots using the following datasets:

- Popularized in Ronald A. Fisher's classic 1936 paper, "The Use of Multiple Measurements in Taxonomic Problems" <https://onlinelibrary.wiley.com/doi/10.1111/j.1469-1809.1936.tb02137.x>
- Data collected by Edgar Anderson on various measurements of 3 species of Iris flowers
- Also found on the UCI Machine Learning Repository. Data available by default on MATLAB

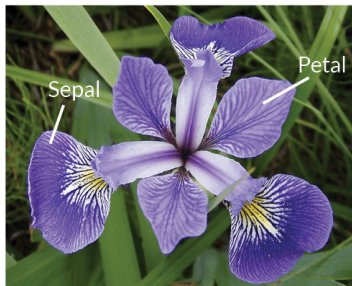
Example 2: Iris species

Example 2: Iris species

Three species of the *Iris* flower:

Example 2: Iris species

Three species of the *Iris* flower:



Iris Versicolor



Iris Setosa



Iris Virginica

Figure: Iris species (Source:

https://s3.amazonaws.com/assets.datacamp.com/blog_assets/Machine+Learning+R/iris-machinelearning.png)

Recap

- Pre-survey review

Recap

- Pre-survey review
- Python/Colab Introduction:

Recap

- Pre-survey review
- Python/Colab Introduction:
 - Summarizing data

Recap

- Pre-survey review
- Python/Colab Introduction:
 - Summarizing data
 - Visualizing data: histograms, boxplots

Problem Sets

Problem Sets

- PS1 due at 1pm today

Problem Sets

- PS1 due at 1pm today
- PS2 will be assigned this afternoon